

WASP-104b

R-2593

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-104_160308 · created 2026-08-27T23:31:33+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2457455.844208 +/- 8.5e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.081 +/- 0.044
Transit depth (Rp/Rs)^2	0.65 +/- 0.7 [%]
Orbital Inclination (inc)	79.2 +/- 2.95
Airmass coefficient 1 (a1)	0.98 +/- 0.57
Airmass coefficient 2 (a2)	0.05 +/- 0.28
Scatter in the residuals of the lightcurve fit is	59.38 %
Best Comparison Star	None
Optimal Aperture	10.98
Optimal Annulus	13.72
Transit Duration (day)	0.017 +/- 0.027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.081 $\pm$ 0.044 vs 0.1214 (deviation 0.85 σ)
Duration fit vs published	0.017 d vs 0.0742 d (deviation 1.97 σ)
Mid-time O-C	-0.4 min
Depth SNR	1.7
Residual scatter	59.38 %

Light curve

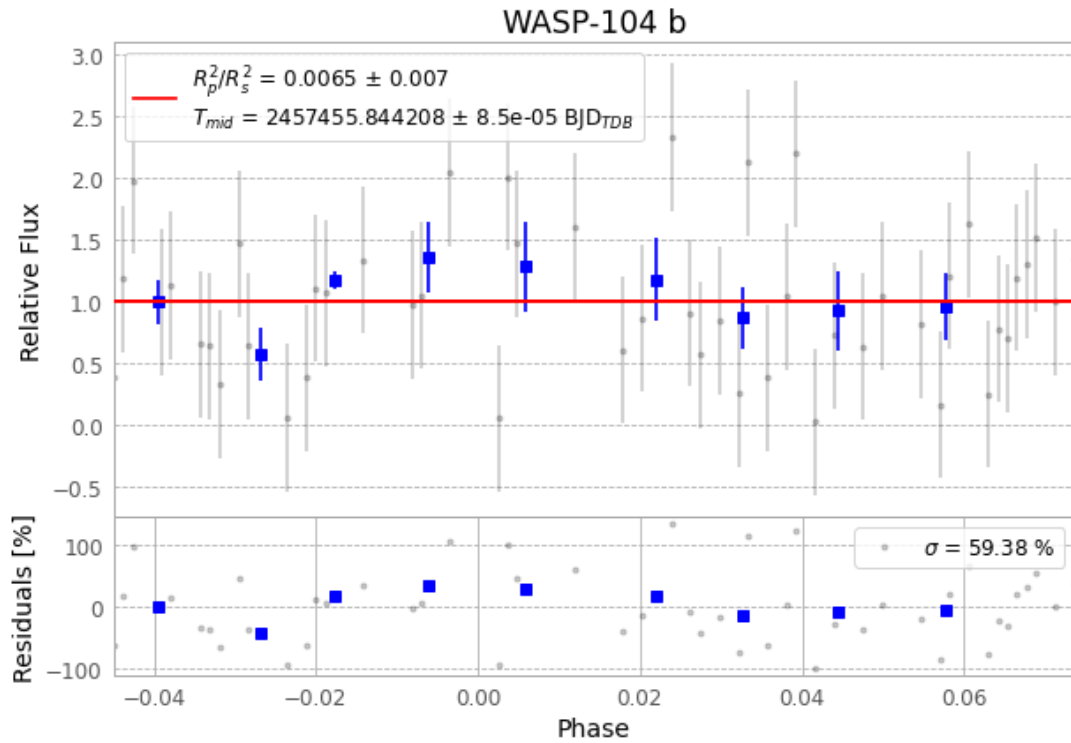


Figure 1 — FinalLightCurve_WASP-104 b_2016-03-07.png (SHA-256 4e9c3e38c840349b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [618, 323], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 64931af04bdc7d88...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

102 FITS frames (67 MB), WASP-104160308061212.FITS → WASP-104160308111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-104-160308.log (4c98e7bc2350...), FinalLightCurve_WASP-104 b_2016-03-07.png (4e9c3e38c840...), FinalLightCurve_WASP-104 b_2016-03-07.csv (5e3793d8a3a5...)

Compute resources

Wall time 158.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-27 23:31Z.